

Stem-and-leaf plots

Lesson 4 of 8



In this lesson you will learn to:

- read values from a stem-and-leaf plot
- put a stem-and-leaf plot in order
- draw a stem-and-leaf plot from a list of data

Name _____

Class _____

Teacher _____

Before you start

You will need

Pencil

Ruler

Syllabus: MA4-DAT-C-01: classifies and displays data using a variety of graphical representations

WE DO

Grey boxes. Work with your teacher. Copy what they write on the board.

YOU DO

Blue boxes. Work on your own. Show your working in the box.

- 1 Work through **Level 1**, then **Level 2**, then **Level 3**. Try the **extension** if you finish early.

- 2 At the end, **show off your skill**: answer one exit ticket question, tear it off and hand it in.

Lesson map

Page	What is on it	Done
3	Learning intentions, key terms, notes and Level 1 teacher examples	☐
4	Level 1: YOU DO practice questions	☐
5	Level 2: WE DO teacher examples and YOU DO practice questions	☐
6	Level 2 and 3: WE DO teacher examples and YOU DO practice questions	☐
7	Level 3: YOU DO practice questions	☐
8	Level 3: YOU DO practice questions	☐
9	Extension questions and the tear-off exit ticket	☐
10	Lesson summary: a worked example for every level	☐

YOU DO

On your own: Warm-up. These skills will help you in this lesson. Show your working.

- 1 Tens digit of 47

- 3 Order: 8, 2, 6, 1

- 2 Units digit of 63

- 4 How many tenths in 0.7?



Acknowledgement of artwork

The artwork used throughout this booklet series was created by **Marni Tuala**. We thank Marni for sharing this artwork with our school community.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
10.04

Stem-and-leaf plots

Date _____

Learning intentions

We are learning to:

- read values from a stem-and-leaf plot
- put a stem-and-leaf plot in order
- draw a stem-and-leaf plot from a list of data

Success criteria

I can:

- ☐ split a number into its **stem** and **leaf** and read the key
- ☐ rewrite an unordered plot as an ordered plot
- ☐ draw an ordered stem-and-leaf plot with a key, including decimals

Key terms Write the definition as your teacher explains it.

Stem-and-leaf plot

Stem

Leaf

Notes

Level 1 · Getting started

Ages of people in a surf class

Stem	Leaf
1	2 5 8
2	0 3 3 7
3	1 4
4	6

Key: 2 | 3 = 23 years

WE DO

With your teacher: copy what they write on the board.

Example 1

Write all the ages in the 2 row.

Example 2

How many people are in the class?
Who is youngest and oldest?

YOU DO

On your own: Set A. Spelling test marks out of 50.

Stem	Leaf
2	4 8
3	0 2 5 5 9
4	1 3 3 3 6 8
5	0

Key: 3 | 5 = 35 marks

- 1 How many students sat the test?
- 2 What was the lowest mark?
- 3 What was the highest mark?
- 4 What is the mode?
- 5 How many scored in the 30s?
- 6 How many scored 40 or more?
- 7 What is the stem of 35?
- 8 What is the leaf of 48?
- 9 How many scored less than 30?
- 10 What number does 4 | 6 stand for?

YOU DO

On your own: Set B. Stems and leaves.

- 1 Stem and leaf of 47?
-

- 4 Stem and leaf of 9?
-

- 2 Stem and leaf of 63?
-

- 5 Order the leaves: 5 | 8 2
-
- 6 1
-

- 3 Stem and leaf of 105?
-

- 6 Order the leaves: 7 | 4 0
-
- 9 3 3
-

- 7 Draw an ordered stem-and-leaf plot of:
-
- 23, 31, 27, 35, 22, 40, 38, 31, 26, 44

Stem	Leaf
2	
3	
4	

Key: ____ | ____ = _____

- 8 How many values are in the 30s?
-

- 9 What is the mode?
-

- 10 What is the highest value?
-

Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 Rewrite this unordered plot as an ordered plot.

Unordered

Stem	Leaf
6	8 2 5
7	3 0 9 1
8	4 4 0

Key: 7 | 3 = 73

Stem	Leaf
6	
7	
8	

Key: ____ | ____ = ____

Example 4 Draw an ordered stem-and-leaf plot of: 56, 43, 61, 48, 55, 59, 62, 47, 55

Stem	Leaf

Key: ____ | ____ = ____

YOU DO On your own: Set C. Masses of rugby players (kg). Questions 3–10 are on the next page.

Unordered plot

Stem	Leaf
7	9 5 8 1 6
8	4 2 5 0 0 8 3
9	0 8 4 2 2 2
10	5 4

Key: 8 | 4 = 84 kg

1 Rewrite the plot as an ordered stem-and-leaf plot.

Stem	Leaf
7	
8	
9	
10	

Key: ____ | ____ = ____

2 How many players are there?

YOU DO On your own: Set C continued. Use the rugby plot on the previous page.

3 What is the lowest mass?

7 How many weigh 90 kg or more?

4 What is the highest mass?

8 What fraction weigh in the 80s?

5 Which mass occurs most often?

9 What percentage weigh 100 kg or more?

6 How many players weigh less than 80 kg?

10 Why is an ordered plot easier to read?

Level 3 · Challenge

WE DO With your teacher: copy what they write on the board.

Example 5 Heights (cm): 142, 155, 138, 149, 151, 160, 147, 138, 153. Use the first two digits as the stem.

Stem	Leaf

Key: ____ | ____ = _____

Example 6 Decimals: 2.4, 3.1, 2.8, 3.5, 4.0, 2.4, 3.9. Stem = whole number, leaf = tenths.

Stem	Leaf

Key: ____ | ____ = _____

YOU DO On your own: Set D. Decimals.

100 m sprint times (seconds): 14.2, 13.8, 15.1, 14.6, 13.5, 14.2, 16.0, 15.4, 14.9, 13.8, 14.2

- 1 Draw an ordered stem-and-leaf plot of the times. Include a key.

Stem	Leaf

Key: ____ | ____ = _____

- 4 What is the mode?

- 2 How many runners were there?

- 3 What was the fastest time?

- 6 What fraction ran 15 seconds or slower?

- 5 How many ran under 14 seconds?

- 7 What is the difference between the fastest and slowest times?

YOU DO On your own: Set D continued. 3-digit numbers.

Scores in a video game: 118, 124, 131, 109, 127, 124, 135, 112, 140

- 8 Draw an ordered stem-and-leaf plot of the scores. Include a key.

Stem	Leaf

Key: ____ | ____ = _____

- 9 Which stems did you use? Why?

- 10 Why must a stem-and-leaf plot have a key?

- 11 Give one advantage of a stem-and-leaf plot over a column graph.

Extension · Stretch yourself

Finished Level 3? Stretch yourself here.

Worked example: split stems

Many values share a stem: 41, 42, 42, 44, 45, 46, 47, 47, 48, 49, 52, 53. Draw a split-stem plot.

- Split each stem into two rows
- Row "4" holds leaves 0 to 4; row "4*" holds leaves 5 to 9
- Do the same for 5 if you need to

Stem	Leaf
4	1 2 2 4
4*	5 6 7 7 8 9
5	2 3

Key: 4* | 7 = 47

Answer: the long row is split, so it is easier to read

YOU DO

On your own: read the worked example, then try these.

- E1** Draw a split-stem plot of: 20, 21, 23, 25, 26, 28, 31, 32, 37

Stem	Leaf
2	
2*	
3	
3*	

Key: ____ | ____ = ____

- E2** A plot has the key 5 | 2 = 0.52. What number does 6 | 7 stand for?

- E3** When is it useful to split the stems?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Unordered plot

Stem	Leaf
3	5 1 8
4	2 0 6 6
5	3

Key: 4 | 2 = 42

Level 1

How many values are in the plot?

Level 2

Write the 4 row with its leaves in order.

Level 3

What fraction of the values are 40 or more? What is the mode?

Lesson summary

Missed the lesson? Start here.

Drawing a stem-and-leaf plot

1. Choose the **stems** (usually the tens) and write them down the left.
2. Write each **leaf** (the units) next to its stem.
3. Rewrite with the leaves **in order**, smallest first.
4. Always add a **key**, e.g. $3 | 5 = 35$.

Level 1

What values are in the 3 row?

1. The stem 3 means 30-something
2. Put each leaf after the stem: 32, 35, 37

Stem	Leaf
2	4 8
3	2 5 7
4	1

Key: $3 | 2 = 32$

Answer: 32, 35 and 37

Level 2

Order this plot: $5 | 7 \ 2 \ 9 \ 6 | 3 \ 0 \ 8 \ 1$

1. Keep the stems
2. Sort each row's leaves from smallest to largest

Stem	Leaf
5	2 7 9
6	0 1 3 8

Key: $5 | 2 = 52$

Answer: $5 | 2 \ 7 \ 9$ and $6 | 0 \ 1 \ 3 \ 8$

Level 3

Draw a plot of 2.3, 3.6, 2.9, 3.1, 4.2

1. Stem = whole number, leaf = tenths
2. Stems 2, 3, 4
3. Order the leaves and add a key

Stem	Leaf
2	3 9
3	1 6
4	2

Key: $2 | 3 = 2.3$

Answer: key $2 | 3 = 2.3$

Extension

Why split a stem into 4 and 4*?

1. A row with lots of leaves is hard to read
2. 4 takes leaves 0–4 and 4* takes 5–9

Stem	Leaf
4	0 2 3
4*	5 7 8 9

Key: $4* | 7 = 4.7$

Answer: it spreads out crowded data

This space is left blank on purpose. It is the back of your exit ticket.